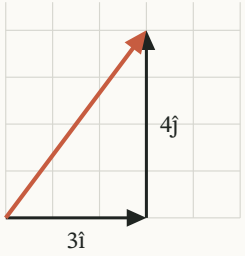
Vectors CHEAT SHEET

THE ARROW



$$\begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$\mathbf{a} = 3\hat{\imath} + 4\hat{\jmath}$

$|\mathbf{a}| = \sqrt{(3^2 + 4^2)} = 5$

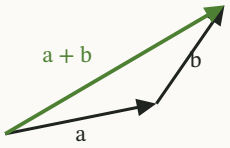
an arrow and its column: the same vector

SCALARS, VECTORS, TYPES

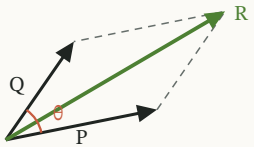
scalar: a size (mass, time, distance)
vector: a size and a direction (displacement, velocity, force)

zero: 0, no direction
equal: same size, same direction
collinear: $\mathbf{b} = k\mathbf{a}$
unit: $|\hat{\mathbf{a}}| = 1$
negative: $-\mathbf{a}$, the other way
coplanar: in one plane

ADDING AND SCALING



triangle law



parallelogram law

$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ $\quad (\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c})$

$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$ $\quad |k\mathbf{a}| = |k|\,|\mathbf{a}|$

$R^2 = P^2 + Q^2 + 2PQ\cos\theta$

$\tan\alpha = \frac{Q\sin\theta}{P + Q\cos\theta}$

θ: the angle between the tails

COMPONENTS

$\mathbf{a} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k} = (a_1, a_2, a_3)$

$|\mathbf{a}| = \sqrt{(a_1^2 + a_2^2 + a_3^2)}$

$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$ $\quad$ (a unit vector: $|\hat{\mathbf{a}}| = 1$)

resolving: $(r\cos\theta, r\sin\theta)$

POINTS IN SPACE

$\mathrm{PQ} = \mathrm{q} - \mathrm{p}$ $\quad$ (head minus tail)

$|\mathrm{PQ}| = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2)}$

$l = \cos\alpha = \frac{a_1}{|\mathbf{a}|}$ $\quad m = \frac{a_2}{|\mathbf{a}|}$ $\quad n = \frac{a_3}{|\mathbf{a}|}$

$l^2 + m^2 + n^2 = 1$

direction ratios: any multiple of (a_1, a_2, a_3)

position vector of P(x, y, z): $x\hat{\imath} + y\hat{\jmath} + z\hat{k}$

FROM THE COURSE: TWO POINTS

P(1, −2, 3), Q(3, 4, 0): $\mathrm{PQ} = (2, 6, -3)$

$|\mathrm{PQ}| = \sqrt{(4 + 36 + 9)} = 7$

unit vector $(2/7, 6/7, -3/7) = (l, m, n)$

$4/49 + 36/49 + 9/49 = 1$

FOUR TESTS

parallel: $\mathbf{b} = k\mathbf{a} \iff \mathbf{a} \times \mathbf{b} = 0$

perpendicular: $\mathbf{a} \cdot \mathbf{b} = 0$

collinear points: $\mathrm{AC} = k\,\mathrm{AB}$

coplanar: $[\mathbf{a}\ \mathbf{b}\ \mathbf{c}] = 0$

SECTION FORMULA

internal, m : n: $\quad r = \frac{m\,b + n\,a}{m + n}$

external, m : n: $\quad r = \frac{m\,b - n\,a}{m - n}$

midpoint: $\frac{a + b}{2}$ $\quad$ centroid: $\frac{a + b + c}{3}$

the weights cross: m goes with b, n with a

FROM THE COURSE: DIVIDING A SEGMENT

A(1, −2, 4), B(7, 4, −2): 1 : 2 inside → (3, 0, 2)

2 : 1 outside → $2b - a = (13, 10, -8)$

midpoint (4, 1, 1); with C(1, 7, 1), centroid (3, 3, 1)

FROM THE COURSE: TWO ROPES ON A RING

6 N and 10 N, 60° apart: $R^2 = 36 + 100 + 60 = 196$

R = 14 N, at about 38.2° to the 6 N rope

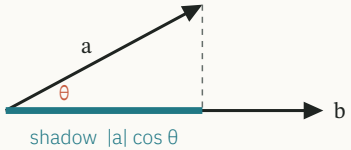
FROM THE COURSE: COMPONENTS

$5\hat{\imath} - 12\hat{\jmath}$: $|\mathbf{a}| = 13$, $\hat{\mathbf{a}} = (5/13, -12/13)$

8 N at 30°: $(4\sqrt{3}, 4) \approx (6.93, 4)$

Vectors CHEAT SHEET

DOT PRODUCT



$a \cdot b = |a|\,|b| \cos\theta$

$a \cdot b = a_1b_1 + a_2b_2 + a_3b_3$

$\cos\theta = \frac{a \cdot b}{|a|\,|b|}$ $\quad a \cdot a = |a|^2$

projection of a on b: $\frac{a \cdot b}{|b|}$ as a vector: $\frac{a \cdot b}{|b|^2}\,b$

$\hat{\imath} \cdot \hat{\imath} = 1,\ \hat{\imath} \cdot \hat{\jmath} = 0$ (and the same for each pair)

$a \cdot b = 0 \iff$ perpendicular

FROM THE COURSE: AN ANGLE, A SHADOW

$(2, -1, 2) \cdot (3, 4, 0) = 2;\ \cos\theta = 2/15;\ \theta \approx 82.3°$

(4, 1, 3) on (2, 1, 2): $15 / 3 = 5$

(2, p, 4) perpendicular to (3, −2, 1): $p = 5$

SCALAR TRIPLE PRODUCT

$[a\ b\ c] = a \cdot (b \times c) = [b\ c\ a] = [c\ a\ b]$

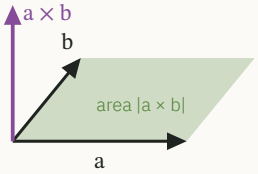
volume of the box $= |[a\ b\ c]|$ $\quad [a\ b\ c] = 0 \iff$ coplanar

PHYSICS

work: $W = F \cdot d = |F|\,|d| \cos\theta$

torque: $\tau = r \times F,\ \ |\tau| = rF \sin\theta$

CROSS PRODUCT



$$a \times b = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

signs + − +

$a \times b = |a|\,|b| \sin\theta\ \hat{n}$

$b \times a = -a \times b$ $\quad a \times a = 0$

$\hat{\imath} \times \hat{\jmath} = \hat{k}$ $\quad \hat{\jmath} \times \hat{k} = \hat{\imath}$ $\quad \hat{k} \times \hat{\imath} = \hat{\jmath}$

the middle term of the determinant takes a minus

AREAS

parallelogram: $|a \times b|$ $\quad$ triangle: $\frac{1}{2}|AB \times AC|$

FROM THE COURSE: A CROSS PRODUCT

$(2, 1, -1) \times (1, 3, 2) = (5, -5, 5)$

checks: $(5, -5, 5) \cdot a = 0,\ \cdot b = 0$

area of the parallelogram $5\sqrt{3} \approx 8.66$

FROM THE COURSE: A BOX, A FLAT BOX

rows (1, 2, 0), (0, 1, 3), (2, 0, 1): volume 13

(1, 2, 3), (2, −1, λ), (3, 1, 4) coplanar: $\lambda = 1$

LINES

$r = a + \lambda b$ $\quad$ through A and B: $r = a + \lambda(b - a)$

$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$ $\quad$ (a, b, c): direction ratios

$\cos\theta = \frac{|b_1 \cdot b_2|}{|b_1|\,|b_2|}$

skew: $d = \frac{|(a_2 - a_1) \cdot (b_1 \times b_2)|}{|b_1 \times b_2|}$

parallel: $d = \frac{|b \times (a_2 - a_1)|}{|b|}$

PLANES

$r \cdot \hat{n} = d$ $\quad$ (d: the distance from O)

$ax + by + cz = d$ $\quad$ normal (a, b, c)

$(r - a) \cdot n = 0$ $\quad$ three points: $n = AB \times AC$

$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 1$ $\quad$ (intercepts p, q, r)

through the line of $P_1 = 0$, $P_2 = 0$: $P_1 + \lambda P_2 = 0$

planes: $\cos\theta = \frac{|n_1 \cdot n_2|}{|n_1|\,|n_2|}$ $\quad$ line, plane: $\sin\varphi = \frac{|b \cdot n|}{|b|\,|n|}$

point to plane: $\frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{(a^2 + b^2 + c^2)}}$ $\quad$ parallel: $\frac{|d_1 - d_2|}{|n|}$

FROM THE COURSE: LINES AND PLANES

skew lines: $d = 6 / (3\sqrt{3}) = 2/\sqrt{3} \approx 1.155$

A(1, 0, 2), B(3, 1, 0), C(0, 2, 1): $3x + 4y + 5z = 13$

(3, 2, 4) to $2x - y + 2z = 6$: $6 / 3 = 2$